Sheela Davi MALHI

AI/ML engineer

Paris, France +33 6 99 93 26 17 sheela-davi.malhi.auditeur@lecnam.net

linkedin.com/in/sheeladavimalhi github.com/davisheela506 Google Scholar ResearchGate

Profile

AI researcher with a Master's in Artificial Intelligence for Connected Industries from CNAM Paris and hands-on experience in edge AI, local LLM inference, experimental evaluation, and real-world AI systems. Completed a seven-month research internship at ITCL Technology Centre, Spain, deploying and evaluating LLM-based agentic systems on an NVIDIA Jetson AGX Orin edge device.

My previous work includes a deep-learning project for diabetic retinopathy detection and co-authored publications on AI in early cancer detection and digital health. Seeking full time opportunities applying my skills to real problems.

Research Interests

Resource-efficient AI, Edge AI, Embedded intelligence, Hardware-aware AI systems, Multi-objective optimization, AI systems evaluation, Sustainable computing, Machine learning, LLMs

Biomedical AI; computational analysis of health-related data; machine learning for biomedical research; microbiome–brain interactions; multi-omics data analysis.

Education

Conservatoire National des Arts et Métiers (CNAM), Paris, France 2024–2026

MSc Artificial Intelligence for Connected Industries (AI4CI) Expected degree: Nov. 2026

- Research-oriented program combining artificial intelligence, networked systems, distributed systems, optimization, embedded systems, reinforcement learning, and AI for networks and IoT.
- Coursework: Machine Learning / AI, Reinforcement Learning, Optimization, Parallel and Distributed Systems, Operating Systems & Computer Architecture, Embedded Systems, Datacenter Design and Operations, AI for Networks & IoT.
- **Thesis: Local agentic AI system for automating business feature extraction and sales workflow integration**

University of Sindh, Sindh University Campus Mirpurkhas, Pakistan 2020–2024

BSc Computer Science 1st position with CGPA: 3.93/4.00

- Final-year project: CNN-based diabetic-retinopathy classification using ImageNet transfer learning and the APTOS 2019 dataset, qualified for NIGRI funding.

Research Experience

ITCL Technology Centre, Burgos, Spain Mar.–Sep. 2026 — GitHub

M2 Research Intern – Agents & Generative AI

- Designed and conducted controlled experiments evaluating five information-extraction systems on real-world business-card data, covering 693 field-level instances.
- Developed an evaluation framework using correct/wrong/missing/invented classifications, field-level accuracy, BERTScore, invalid-output rate, and reliability analysis.
- Benchmarked Qwen3.6, Gemma 4, Ministral 3, and a DeepSeek OCR + rule-based baseline; the strongest model achieved 88.6% field-level accuracy.
- Deployed local LLM inference on an NVIDIA Jetson AGX Orin and monitored computational constraints including GPU/CPU utilisation, memory usage, and inference latency.
- Analysed performance by field type and proposed field-specific post-processing and reasoning strategies to improve reliability.

Methods/tools: Python, Pandas, model benchmarking, quantitative evaluation, local LLM inference, Docker/Linux

CEDRIC, CNAM, Paris, France Jun.–Sep. 2025

M1 Research Intern – ML and Generative-AI-Powered Insights from 5G Traffic

- Worked with large-scale real-world mobile-network traffic data (tens of gigabytes; hundreds of millions of records) and developed preprocessing workflows for heterogeneous temporal data.
- Investigated abnormal traffic patterns using machine-learning methods and Python data-analysis tools.
- Contributed to a monitoring pipeline connecting Kafka event streaming, PostgreSQL and Grafana for data handling and visual monitoring.
- Worked within practical memory constraints by processing data in chunks and structuring data preparation for

subsequent analysis.

Methods/tools: Python, Pandas, scikit-learn, TensorFlow, Jupyter, Kafka, PostgreSQL, Grafana

Research Projects

CarbonNAS — Self-Directed Research project in Resource-Efficient AI Oct. 2026 — GitHub

- Developing a PyTorch-based neural architecture search (NAS) framework for exploring accuracy–efficiency trade-offs in resource-constrained AI models.
- Implemented a configurable CNN search space with varying network depth, channel configurations, kernel sizes, and pooling strategies, and automated candidate architecture generation and evaluation.
- Evaluated candidate architectures using parameter count, FLOPs, model size, and inference latency; observed up to **59× variation in FLOPs** and **11× variation in latency** across sampled architectures.
- Extending the framework toward multi-objective optimization with accuracy, energy consumption, and carbon intensity for carbon-aware model selection.

Technologies: Python, PyTorch, Torchvision, NumPy, Pandas, THOP, Git/GitHub, CIFAR-10

Automated Diabetic Retinopathy Detection 2023–2024

- Developed a CNN-based image-classification system using ImageNet transfer learning on the **APTOS 2019** dataset for five-class diabetic retinopathy classification.
- Applied data augmentation and regularisation to improve model robustness; achieved 90% accuracy on the test set.
- Built a Flask and MongoDB-backed prototype for model inference and result handling.

Technologies: Python, TensorFlow, CNNs, transfer learning, medical image analysis, Flask, MongoDB.

Smart Bio-Reactor Monitoring and Forecasting Platform 2024–2025

- Designed a monitoring workflow combining streaming data, database storage and Grafana visualisation to support observation of system behaviour over time.
- Explored LSTM-based time-series modelling and Prophet-based forecasting/anomaly detection as complementary approaches to identifying changing patterns.
- Used the project to connect data ingestion, time-series analysis and visual monitoring in an applied systems context.

Methods/tools: Python, Kafka, PostgreSQL, Grafana, LSTM, Prophet.

Publications

- **Artificial Intelligence in Early Cancer Detection: A Paradigm Shift in Oncology Diagnostic.** I. Essa, **S. D. Malhi**, D. K. Malhi, T. Asghar, M. Tahir, S. K. Muhammad. *Journal of Medical & Health Sciences Review*, 2(3), 2025.
- **Perceived Impact of Artificial Intelligence Usage on Academic Performance Among Undergraduate Health Science Students in Mirpurkhas, Sindh-Pakistan: A Multi-Center Study.** S. K. Ahmed, I. A. Chandio, A. Pervaiz, A. Fatima, **S. D. Malhi**. *Mader-e-Milat International Journal of Nursing and Allied Sciences*, 3(4), 47–60, 2025.
- **Digital Health Knowledge, Attitudes, and Practices Among Undergraduate Nursing Students.** I. A. Arain, I. A. Chandio, A. Z. Rajput, N. R. Gill, **S. D. Malhi**, D. K. Malhi. *Link Medical Journal of Health and Community Research*, e90, 2025.
- **Integrating Artificial Intelligence to Foster Critical Thinking and Problem-Solving in 21st-Century Classrooms.** S. A. Shahzad, M. M. Raza, **S. D. Malhi**, Q. ul Ain, D. K. Malhi. *Research Journal for Social Affairs*, 3(5), 635–648, 2025.
- **Awareness and Attitudes Towards Artificial Intelligence-Generated Study Aids Among Nursing Students in Public and Private Nursing Colleges of Mirpurkhas, Sindh, Pakistan.** **S. D. Malhi** *NURSEARCHER: Journal of Nursing & Midwifery Sciences*, 2026.

Google Scholar: 20 citations · h-index: 2 · i10-index: 1

Teaching & Academic Experience

University of Sindh Jan.–May 2024

Teaching Assistant – Artificial Intelligence With Prof. Sarwat Nizamani

- Supported teaching and practical learning activities in Artificial Intelligence, Guided 50+ students in supervised and unsupervised machine learning and explained neural networks. and optimisation techniques

Technical Skills and Certifications

Machine Learning: Python, PyTorch, TensorFlow, Scikit-learn, CNNs, transfer learning, anomaly detection, model evaluation, neural architecture search

LLM / Agentic AI: LLM evaluation, agentic architectures, OpenClaw, local LLM inference, tool use, structured-output evaluation, hallucination and failure-mode analysis

Data / Systems: Pandas, Jupyter, MongoDB, Flask, Docker, Linux, distributed systems, network traffic analysis, 5G data processing

Research: Experimental design, benchmarking, quantitative analysis, data preprocessing, literature review, scientific writing, interpretation of results, interdisciplinary collaboration.

Certifications: Data Science – Johns Hopkins University; Google Data Analytics; IBM AI Foundations for Business; Introduction to Data Science; AI Foundations for Everyone; Machine Learning – Stanford University.

Languages

English: B2 / professional (Academic work, master's thesis, publications, and research internships conducted in English)
French: A2 **Urdu/Sindhi:** Native

Awards & Honors

- **Vice Chancellor's Silver Medal**, University of Sindh, 2024 – Awarded for first position / top academic performance in the BSc Computer Science program.
- **PM Youth Laptop Program**, Government of Pakistan, 2024 – Awarded based on academic merit.
- **Erasmus+ Mobility Grant**, European Union, 2026 – Awarded for research internship mobility at ITCL Technology Centre, Burgos, Spain.

References

Prof. Stefano Secci, Professor of Communication Networks, CNAM Paris (stefano.secci@lecnam.net)
Prof. BRACONNOT-VELLOSO Pedro, Associate Professor, CNAM (pedro.velloso@lecnam.net)
Mr. Marteyn VAN GASTEREN, R&I Manager International Projects, ITCL (marteyn.vangasteren@itcl.es)